

Subject Description Form

Subject Code	EIE4432
Subject Title	Web Systems and Technologies
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	ENG2002 Computer Programming AND ENG2003 Information Technology
Objectives	This subject will provide students with the principles and practical programming skills of developing Internet and Web applications. It enables students to master the development skill for both client-side and server-side programming, especially for database applications. Students will have opportunity to put into practice the concepts through programming exercises based on various components of client/server web programming.
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: <u>Category A: Professional/academic knowledge and skills</u> 1. Understand the enabling technologies for building Internet and Web database applications. 2. Understand the different components for developing client/server applications. 3. Apply the techniques and features of the client/server development languages to construct a database application based on Internet. 4. Develop the web database applications through programming exercises. <u>Category B: Attributes for all-roundedness</u> 5. Present ideas and findings effectively. 6. Think critically. 7. Learn independently.
Subject Synopsis/ Indicative Syllabus	Syllabus: 1. <u>Introduction to Client/Server Computing</u> The basic principles of client/server computing; Distinguished characteristics of client/server systems and application areas; Comparison of two-tier versus three-tier client/server solutions; Web programming model; Interactive web. 2. <u>Client-Side Web Programming</u> Benefits and limitations of client-side web programming. Basic concepts and development based on HTML, CSS, JavaScript, jQuery, AJAX, JSON. 3. <u>Server-Side Web Programming</u> Approaches to server-side programming. Benefits and limitations of server-side web programming. Development framework for server-side programming based on Node.js, RESTful API. Web application development: Development of a web application; Comparison of synchronous and asynchronous techniques.

[illegible]

	The continuous assessment consists of quiz(zes), test(s), laboratory exercises, and project. Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: <table border="1" data-bbox="400 318 1465 763"> <tr> <th data-bbox="400 318 715 398">Specific Assessment Methods/Tasks</th><th data-bbox="715 318 1465 398">Remark</th></tr> <tr> <td data-bbox="400 398 715 636">Tests, quizzes</td><td data-bbox="715 398 1465 636">End-of-chapter type problems used to evaluate students' ability in applying concepts and skills learnt in the classroom; students need to think critically and creatively in order to come up with an alternate solution for an existing problem.</td></tr> <tr> <td data-bbox="400 636 715 763">Laboratory sessions, Project</td><td data-bbox="715 636 1465 763">The laboratory exercises and the project are to evaluate students' technical knowledge and communication skills.</td></tr> </table>	Specific Assessment Methods/Tasks	Remark	Tests, quizzes	End-of-chapter type problems used to evaluate students' ability in applying concepts and skills learnt in the classroom; students need to think critically and creatively in order to come up with an alternate solution for an existing problem.	Laboratory sessions, Project	The laboratory exercises and the project are to evaluate students' technical knowledge and communication skills.								
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Reading List and References	Reference Books: 1. V. Subramanian, <i>Pro MERN stack : full stack web app development with Mongo, Express, React, and Node</i>, Second edition. New York, NY: Apress, 2019. 2. C. Northwood, <i>The Full Stack Developer: Your Essential Guide to the Everyday Skills Expected of a Modern Full Stack Web Developer</i>. Berkeley, CA: Apress L. P, 2018. 3. Rungta, Krishna. <i>Learn NodeJS in 1 Day: Complete Node JS Guide with Examples</i>. Independently published, 2016. 4. V. Bojinov and V. Bojinov, <i>RESTful web API design with Node.js 10 : learn to create robust RESTful web services with Node.js, MongoDB, and Express.js</i>, Third edition. Birmingham: Packt Publishing Ltd., 2018. 5. S. Powers, <i>JavaScript cookbook : programming the web</i>, Third edition. Sebastopol, California: O'Reilly Media, Incorporated, 2021.														